

UNIT 7

SUSTAINABLE COMPUTING



Connect to the topic

1. Look at the photo.
What do you think happened to these computers and gadgets?
2. Why do people throw away electronics?
3. What problems can e-waste cause?
4. Do you think companies should make computers that last longer?
5. What are some eco-friendly ways to dispose of electronics?

WARM-UP VIDEO

WATCH AND SPECULATE. Watch the video about green computing and discuss the questions below.

1. Why is green computing important?
2. How do data centers contribute to environmental problems?
3. How can energy-efficient hardware help the environment?
4. What is virtualization, and how does it support green computing?
5. What small actions can individuals take to support green computing?
6. Could green computing slow down technological progress? Why or why not?
7. Should tech companies be responsible for recycling old devices?



UNIT 7.1 E-WASTE AND RECYCLING

LISTENING

Task 1. EXPLORE THE WORDS. Match the words to their definitions.

1. equate	a) to throw something away
2. substantial	b) to fail to reach a target or expectation
3. comprise	c) knowledge or understanding of a situation or issue
4. survey	d) to consist of or include
5. discard	e) large in amount or importance
6. retain	f) to be equal to something in amount or value
7. fall short	g) to keep or hold onto something
8. awareness	h) a study where people are asked questions to collect information

Task 2. LISTEN FOR DETAILS. Scan the QR code and listen to the audio report “Old Electronic Devices in the EU”. Match each sentence to the correct figure mentioned in the listening. There are extra figures you do not need to use.



1. The amount of e-waste collected per person in the EU in 2022 (in kg).	11.2
2. The percentage of EU citizens who kept their old mobile phones at home.	40.6
3. The percentage of people who recycled their old laptops or tablets.	51
4. The percentage of individuals who recycled their old desktop computers.	10
5. The overall WEEE collection rate in the EU in 2022.	2022
	6
	5 million
	13

Task 3. LISTEN FOR DETAILS. Listen to the report again. Decide whether the following statements are True, False, or Not Given.

1. The EU achieved its 65% target for WEEE collection in 2022.	True / False / Not Given
2. Sweden had the highest rate of individuals recycling old desktop computers.	True / False / Not Given
3. Most EU citizens throw away their old mobile phones without recycling.	True / False / Not Given
4. The WEEE Directive aims to improve the environmental performance of electronic device producers.	True / False / Not Given
5. In 2022, 14.4 million tons of electrical and electronic equipment were placed on the EU market.	True / False / Not Given
6. The recycling rate for laptops and tablets in the EU was higher than that for desktop computers.	True / False / Not Given
7. The EU plans to ban the export of e-waste to non-EU countries by 2028.	True / False / Not Given
8. 1% of EU citizen discarded old mobile phones without recycling.	True / False / Not Given

READING AND VOCABULARY

Task 4. READ FOR MAIN IDEAS. Read the text about handling e-waste. Find out which speaker best matches the following statements. One of the statements doesn't relate to the text and should be marked 'not given'.

1. We want customers to have easy access to spare parts.
2. We believe phones should be made from biodegradable materials.
3. We work with a team that recycles metals from phones.
4. We help reduce the number of working devices going to landfills.
5. We believe repairing is better than replacing devices.
6. We want to make sure apps work well on older devices.

How Tech Companies Handle E-Waste

Lucas, 29

I work in logistics at a large tech company. Previously, we would dispose of enormous quantities of returned and unsold electronics. Now we sort them: reusable, recyclable, and hazardous. Those that are still in good condition are sent to refurbishment centers. We refurbish and clean them, then sell them at a lower price or donate them to charities and schools. This way, there are fewer phones that end up in landfills. I think this program is not only sustainable but also good for our company image.

Isabella, 42

I'm an environmental manager at a phone manufacturer. We've started a take-back program. Customers can bring used phones to our stores, and we recycle the components. The rare metals in phones are valuable but tricky to safely recycle. We work with approved recycling partners to make sure the materials are recovered properly. Some parts, like batteries, must be specially handled due to toxic chemicals. Our biggest challenge is to get more people to return devices. Only a few do.

Mateo, 37

I'm a software engineer and don't usually work on hardware. But I've just begun working on a project that decreases e-waste through software updates. Most people upgrade to new devices just because their old devices become slow or unsupported. Our idea is to keep products alive longer by providing long-term software updates. I think it's a smart way of producing more sustainable devices. We also test our apps on old phones to make sure they still perform well. It's a small step, but it can make a tremendous difference.

Lena, 34

I'm the CEO of a small startup that makes modular laptops. What that means is that customers can replace parts like the battery, screen, or keyboard easily without having to throw away the whole device. We're strong believers in the right to repair. If something's broken, people should be able to fix it instead of buying new. Our laptops come with manuals, and we sell replacement parts online. It's challenging from a design perspective, but our customers appreciate the flexibility and the sustainability.

Task 5. READ FOR DETAILS. Read the text again. Choose the correct answer.

1. What happens to devices that Lucas's company can still use?
 - A. They are thrown away.
 - B. They are sold at higher prices.
 - C. They are repaired and reused.
 - D. They are melted down into raw materials.
2. What is the goal of Isabella's take-back program?
 - A. To encourage phone upgrades.
 - B. To design phones with cheaper parts.
 - C. To collect and recycle old devices safely.

- D. To reduce battery size.
- 3. What issue does Mateo's team try to solve?
 - A. Slow charging speeds.
 - B. High device prices.
 - C. Defective batteries.
 - D. Short software support.
- 4. What is a feature of Lena's laptops?
 - A. Their parts can be replaced by the users.
 - B. They must be sent to the company for repairs.
 - C. They are sold only to eco-friendly companies.
 - D. They are made of biodegradable materials.
- 5. What does Isabella mention as a major difficulty in their recycling efforts?
 - A. Finding certified recycling partners
 - B. Making phones easier to take apart
 - C. Transporting old phones back to factories
 - D. Getting customers to return old devices
- 6. Why does Mateo believe software updates are important for sustainability?
 - A. They reduce electricity consumption.
 - B. They allow devices to work longer without needing replacement.
 - C. They help recycle hardware components.
 - D. They make devices charge faster.

Task 6. COMMUNICATE. Work in pairs. Discuss the questions below.

- 1. Have you ever returned or recycled an electronic device? What happened?
- 2. Would you prefer to repair your device or buy a new one? Why?
- 3. Do you think tech companies should be responsible for collecting and recycling old products?
- 4. What could motivate more people to return their used electronics?
- 5. How could software updates help reduce e-waste?

Task 7. WORK WITH WORDS. Fill in the text with the words from the box.

Extending the Life of Electronics

refurbished ▪ collection point ▪ biodegradable ▪ disassembly ▪ scarcity ▪
take-back ▪ obsolescence ▪ cradle-to-cradle ▪ repairability ▪ upgradable ▪

In recent years, more companies and consumers have started to rethink how we treat old electronic devices. Instead of throwing them away, people are now encouraged to bring them to a 1) _____, where they can be sorted for 2) _____ or repair. This process allows useful components to be reused and materials to be recovered.

Some electronics are sent to be 3) _____ – cleaned, repaired, and updated for resale. This is especially important in a time of material 4) _____, where resources like rare earth metals are becoming harder to find and extract. Designers are also creating more 5) _____ devices so that parts can be replaced without needing to buy a whole new product.

In addition, 6) _____ schemes are becoming common. These programs allow consumers to return their old gadgets to the manufacturer for recycling or repair. This approach supports the idea of 7) _____ design, where products are made to be reused or remanufactured, not simply discarded.

Reducing planned 8) _____, when devices are designed to fail or become outdated quickly, is also key. Companies are now rated on the 9) _____ of their products, making it easier for customers to choose sustainable options.

Some businesses are also experimenting with 10) _____ materials in electronics, especially for packaging or small devices, to reduce the long-term impact on the environment.

Task 8. COLLABORATE. Work in pairs. Look at a list of actions, technologies, and programs related to e-waste. Place each item in the correct column of the table.

Prevention (Reducing waste before it's created)	Recycling & Recovery (Processing old devices/materials)	Disposal (End-of-life, non-recoverable actions)

- Designing upgradable smartphones
- Repair cafés for broken electronics
- Throwing a laptop in household waste
- Extracting copper
- Collection point at electronics store
- Burning wires to extract metals
- Take-back schemes by manufacturers
- Throwing away discarded appliances
- Software updates to extend device use
- Refurbishing old tablets for resale
- Disassembling phones to recover parts
- Burning non-functional plastic cases

WATCHING

Task 9. EXPLORE THE WORDS. Complete the definitions with the phrases below.

- | | | |
|---------------------------------|----------------------------|-----------------------|
| a) thrown away | b) buried under the ground | c) can be extracted |
| d) quickly or with force | e) such as chemicals | f) an action or event |
| g) a toxic metal that can cause | h) valuable materials | i) waste materials |
| | j) excellent conductivity | |

1. **urban mining** (n): the process of recovering _____ from electronic waste found in cities
2. **discarded electronics** (n): electronic devices that have been _____.
3. **ore** (n): natural rock or sediment containing valuable minerals that _____.
4. **copper** (n): a reddish-brown metal used in electronics for its _____.
5. **pluck out** (v): to pull something out _____, often with the fingers.
6. **landfill** (n): a place where waste is _____, often used for general garbage disposal.
7. **ramification** (n): a possible consequence of _____, often negative or complex.
8. **lead exposure** (n): the condition of being in contact with lead, _____ serious health problems.
9. **susceptible** (adj): likely to be harmed by something, _____.
10. **incinerator** (n): a machine or facility used to burn _____ at high temperatures.

Task 10. WATCH FOR DETAILS. Scan the QR code and watch the video “E-waste”, then choose the correct answers to the questions.



1. Why is Ghana shown in the video?
 - A. It produces many electronics.
 - B. It recycles phones in factories.
 - C. It receives large amounts of e-waste.
 - D. It has banned phone imports.
2. Why is working with e-waste dangerous?
 - A. It can break the machines.
 - B. It's expensive to recycle.
 - C. It creates harmful smoke and chemicals.
 - D. It makes electronics stop working.
3. Why do people burn e-waste?
 - A. To get valuable metals.
 - B. To cook food.
 - C. To remove plastic.
 - D. To stop pollution.
4. Why does the chemist care about the problem?
 - A. Her lab is next to a landfill.
 - B. She works for Fairphone.
 - C. She is a scientist and a mother.
 - D. She buys phones in Ghana.
5. What health problem did the chemist find?
 - A. Phones cause hearing loss.
 - B. Chemicals were in breast milk.
 - C. Food smells like smoke.
 - D. E-waste makes babies cry.
6. What did the video say about gold in smartphones?
 - A. They have no gold.
 - B. There is more gold in phones than in gold ore.
 - C. One phone equals one gold ring.
 - D. Only old phones have gold.
7. Why is Fairphone easier to recycle?
 - A. It's the cheapest phone.
 - B. It breaks easily.
 - C. It uses only plastic.
 - D. It has simple parts that are easy to replace.
8. What problem does the Fairphone creator talk about?
 - A. Electronics are too fast.
 - B. Phones are hard to repair or reuse.
 - C. Phones are too heavy.
 - D. Phones don't have enough features.
9. What does the entrepreneur say about how products are made?
 - A. They are designed to be used for many years.
 - B. They are designed to break quickly.
 - C. They are easy to make at home.
 - D. They always go back to the factory.
10. What does the video say about rich countries?
 - A. They recycle everything.

- B. They don't throw away electronics.
- C. They create most of the world's e-waste.
- D. They help clean up e-waste in Africa.

Task 11. WORK WITH WORDS. Match the words from two columns to make a collocation from the video.

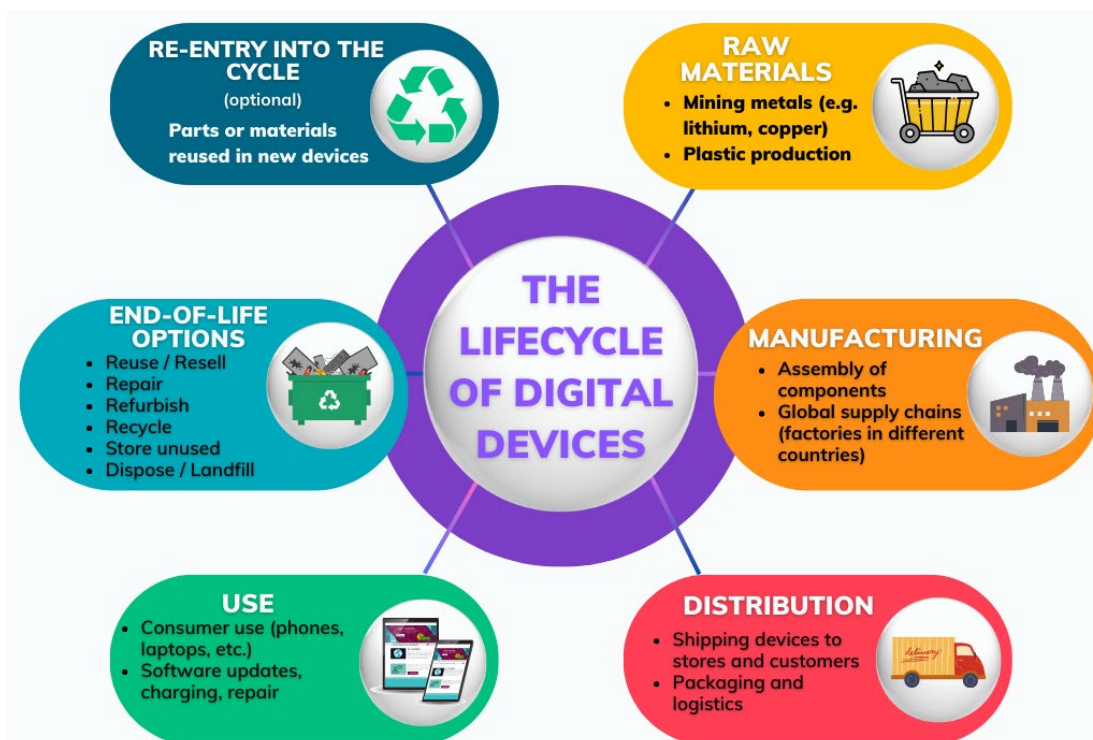
1. circuit	a) equipment
2. safety	b) to chemicals
3. live	c) modular parts
4. extract	d) board
5. set	e) warranty
6. susceptible	f) minerals
7. sustainable	g) of concept
8. removable	h) a fire
9. proof	i) phone
10. under	j) on the margins

Task 12. COMMUNICATE. Work in pairs or small groups. Discuss the following questions. Try to support your ideas with examples or personal opinions.

1. What surprised you the most about the video? Why?
2. Did you know what happens to your old electronics after you throw them away or recycle them? What do you think should happen?
3. Do you agree that we focus more on where products come from than where they go after use? Why do you think that is?
4. In the video, urban mining is described as both dangerous and necessary. Should poor communities continue to do this work? Why or why not?
5. Is it fair that the consequences of our electronic waste often affect people in countries like Ghana?
6. The journalist calls the system of burning electronics to extract materials "a stupid process." Do you agree? Why or why not? What could be a smarter process?
7. Would you be willing to pay more for a device like the Fairphone, which is easier to repair and better for the environment? Why or why not?
8. In the video, a scientist found toxic chemicals in breast milk from women living near e-waste sites. How should governments respond to this kind of health risk?
9. Many people replace their phones every 2–3 years. What are the real-life consequences of this trend? Should phone makers be responsible?
10. What can individuals, like students or consumers, do to reduce the impact of e-waste?

SPEAKING

Task 13. COLLABORATE. Work in pairs. Look at each stage of the digital device lifecycle. Discuss what risks or problems can happen at each stage.



Choose one or two stages from the lifecycle. Brainstorm realistic ways to make them more sustainable.

- Which stage is the most and the least damaging to the environment?
- Which stage is hardest to change?
- Which is easiest?
- Where can individuals make the biggest difference?

Task 14. COLLABORATE. Work in small groups. Read the descriptions of different countries' e-waste laws. Put them into 3 categories:

- Strong e-waste laws
- Moderate and developing laws
- Weak or informal systems

European Union	Follows the WEEE Directive, which requires producers to take back, recycle, or properly dispose of e-waste. There are clear recycling targets and rules for labeling products.
Switzerland	One of the first countries to introduce a national recycling program for electronics. Consumers can return old devices at collection points without extra cost.
Ghana	Many electronics are shipped here from richer countries. Much of the e-waste is processed in dangerous, informal ways by people who collect parts for money.
United States	There is no national e-waste law, but some states (like California and New York) have their own recycling programs. In other states, electronics can still be thrown away in the trash.
Japan	Uses the Home Appliance Recycling Law. When you buy new electronics, you also pay a recycling fee. This helps cover the cost of proper disposal.
India	Has e-waste rules that make producers responsible for collecting and recycling electronics. The system is still developing, and many devices are still processed informally.
China	Banned imports of foreign e-waste and created its own recycling programs. But

	illegal imports and informal recycling still happen in some areas.
Brazil	E-waste recycling laws exist, but enforcement is weak. Most e-waste ends up in landfills or is recycled by informal workers without safety measures.
Norway	Has one of the highest e-waste recycling rates in the world. Its system includes take-back points and strong cooperation between producers and recyclers.
Australia	The National Television and Computer Recycling Scheme help people return e-waste for free. However, many people still store or throw away their old electronics.

Task 15. COLLABORATE. Work in small groups. Rank these actions from 1 (most effective) to 8 (least effective) for reducing e-waste globally. You must agree as a group and explain reason why.

- Offer take-back schemes for old electronics.
- Make devices more upgradable (e.g. replaceable parts, longer software support).
- Ban planned obsolescence (making products break or slow down on purpose).
- Add e-waste education to schools and public campaigns.
- Give tax discounts for using refurbished devices.
- Support repair cafés or local repair shops.
- Ban e-waste exports to poorer countries.
- Use biodegradable materials in electronic production.

Task 16. COMMUNICATE. Work in pairs. Discuss what can happen if someone doesn't wipe data from their old phone, laptop, or hard drive before selling or recycling it. Answer the questions below.

1. What types of personal data are stored on devices?
2. What risks could happen if a stranger gets access to this data?
3. Who do you think is most vulnerable – teens, adults, companies?

Useful Phrases:

If you don't wipe your data, someone might...

One possible risk is...

To protect your privacy, you should always...

Task 17. COMMUNICATE. Work in pairs. Discuss which devices are most dangerous if not wiped. Which devices store the most private data? Which are easiest to forget about?

- Personal smartphone
- Office desktop computer
- Family tablet
- External USB drive
- Smart watch

Discuss how to wipe data from a device before giving it away or recycling it. Answer the questions below.

1. Do you usually delete files, or fully reset your device?
2. Have you ever used data-wiping software?
3. Do you think deleting is enough?
4. Would you help a friend wipe their device safely?

Task 18. COLLABORATE. Work in small groups. Create a guide "How to Wipe Your Phone Before Recycling It". Include:

- tools or settings to use
- what to do with SIM cards and SD cards

- final check before giving the device away
- Present your guide to the class.

Task 19. ANALYZE. Read the case study and discuss possible solutions to the problem.

A well-known electronics brand had just rolled out a new recycling program “Trade in your old device, get a discount voucher”. It seemed like a smart move. They had posters everywhere and ads ran online for weeks. The company was ready to collect tons of old tech. But when the first results came in, the numbers were surprisingly low. Only around 8% of customers had brought anything back. No one could quite explain why it was so. Later, follow-up surveys revealed something more basic: most people didn’t actually know what counted as e-waste. Some didn’t realize old chargers or broken phones were recyclable. Others simply didn’t see them as harmful, just clutter in a drawer. The company now faces a different kind of challenge. It’s not just about giving people discounts, it’s about helping them understand why recycling electronics matters in the first place.

What kinds of changes might actually get more people to return their used devices instead of tossing them?

LANGUAGE FOCUS

Task 20. STUDY AND ANALYZE. Look at the rule about clauses of result, study in what situations they are used.

CLAUSES OF RESULT

WORDS/EXPRESSIONS	EXAMPLE
Such a(n) + (adjective) + singular countable noun	It was such an efficient recycling program (that) we reduced e-waste by 30%.
Such + (adjective) + uncountable/plural noun	There was such heavy pollution (that) the surrounding areas became hazardous.
such a lot of + plural or uncountable nouns	The program collected such a lot of used batteries (that) it took weeks to process them all.
So + adjective/adverb	The materials were so rare (that) they were sold to manufacturers for reuse.
so much/little + uncountable so many/few + plural noun	There is so much electronic waste (that) recycling centers struggle to keep up. The computers have so many rare metals (that) they can be recycled for profit.
As a result / therefore / consequently	The e-waste was not properly disposed of, and as a result , it harmed the environment. There was a delay in collection; therefore , the recycling process was slow. The program increased awareness, consequently , more people started recycling their old electronics.

Task 21. PRACTICE. Match the two parts of a sentence.

1. Many people throw away old electronics,	a) as a result, companies must recycle properly.
2. The process was so complex that	b) that many companies started using it.
3. Factories create so much toxic waste that	c) as a result, landfills are filling up.

4. The company started a buyback program,	d) that they throw away useful materials.
5. People know so little about e-waste	e) it harms the environment.
6. The government made strict e-waste rules,	f) it took hours to complete the disassembly.
7. Companies replace so few parts	g) therefore, more people recycle their devices.
8. It was such an innovative recycling program	h) that people must buy new devices.

Task 22. PRACTICE. Fill in the gaps with so, such and such a.

- A: The recycling event had _____ many participants that they ran out of collection bins.
B: That's great! It shows people care about the environment.
- A: Why are you in _____ hurry?
B: Because I have to drop off my old laptop at the recycling center before it closes.
- A: I have _____ much e-waste at home! I don't know what to do with it.
B: You can take it to a collection point for proper disposal.
- A: It's _____ nice to see people recycling their old devices.
B: Yes, it helps reduce pollution.
- A: What did you think of the new recycling program?
B: I loved it. I've never seen _____ well-organized initiative before.
- A: I want to buy a new laptop, but this one still works.
B: That's true. It would be _____ wasteful to throw it away.
- A: There was _____ little awareness about e-waste in the past.
B: Yes, but now more people know how to recycle.
- A: The old computer contained _____ valuable metals that it was worth recycling.
B: That's why we should never throw electronics in the trash.
- A: This battery is leaking.
B: Be careful! It contains _____ dangerous chemicals.
- A: The factory produces _____ much toxic waste.
B: They need better recycling processes.
- A: There was _____ lot of e-waste dumped in landfills.
B: That's why we need better recycling programs.
- A: I was shocked to see _____ many discarded gadgets at the landfill.
B: Yes, e-waste is a big problem worldwide.

UNIT 7.2 ENERGY-EFFICIENT COMPUTING

READING AND VOCABULARY

Task 23. READ FOR DETAILS. Read the text and choose the correct ending for a sentence.

Smarter Tech for a Greener Future

The world of computers, servers, and networks today depends on powerful computers, servers, and networks. But all that hardware takes lots of electricity, especially in large data centers that do not go offline. With more and more individuals streaming movies online, storing files in the cloud, and working online, energy consumption continues to increase. That produces higher carbon emissions and leads to climate change.

That is why more research is being concentrated on energy-efficient computing. It targets reducing the power consumption of IT systems without impacting their speed and reliability. One method of saving power is to reduce the power requirement of systems. Low-power processors are employed to this purpose. They carry out usual activities such as browsing, video chatting, or office applications, just at reduced power and heat. New laptops and tablets now include these power-saving chips in order to extend battery life and produce less heat.

Another solution is better cooling systems in data centers. Instead of using air conditioning all the time, companies now use liquid cooling, natural airflow, or smart temperature controls. Google, for example, uses AI-powered software to manage its server energy use. This kind of automated energy optimization helps reduce waste.

Virtualization also helps. It allows multiple systems to run on one physical machine, which lowers the need for hardware and cuts power use. Many companies now use cloud computing to store data and run software more efficiently.

Even individuals can help. Enabling sleep mode, reducing screen brightness, and turning off unused devices are simple steps. Choosing Energy Star–certified devices or eco-mode settings on printers and monitors also supports a greener lifestyle.

In short, energy-efficient computing is about using smarter design, better software, and greener habits. It's good for the planet and good for your electricity bill, too.

(adapted from <https://info.pivotalglobal.com/blog/data-center-energy-consumption-environmental-impact>)

1. Data centers use a lot of electricity because they run *only at night* / *all the time* / *only for streaming services*.
2. Energy-efficient computing helps by *using stronger energy* / *making devices look better* / *lowering energy use*.
3. To reduce heat in data centers, companies now use *cold weather storage* / *better cooling systems like liquid cooling and airflow* / *smaller energy*.
4. Google uses AI-powered software to *train servers* / *clean data centers* / *manage server energy use*.
5. Virtualization helps save energy by *using one machine for many systems* / *using many machines* / *switching off the internet*.
6. Cloud computing helps companies *store more physical hardware* / *reduce energy and hardware* / *use less paper documents*.
7. People can save energy at home by *playing less games* / *turning off devices not in use* / *charging devices all night*.
8. Energy Star–certified products *use more power to work faster* / *are cheaper than others* / *use less energy*.

Task 24. WORK WITH WORDS. Match the words to their definitions.

1. carbon emissions	a) to make something last longer
2. power consumption	b) using liquid (usually water) to keep a computer from overheating

3. extend	c) controlled by artificial intelligence
4. liquid cooling	d) using the internet to store data or run apps
5. natural airflow	e) low-power state when not in use
6. AI-powered	f) harmful gases released into the air
7. virtualization	g) officially marked as energy-saving
8. cloud computing	h) the amount of electricity a device or system uses
9. sleep mode	i) running many systems on one computer
10. Energy Star-certified	j) using air from outside or smart ventilation to cool equipment.

Task 25. EXPLORE THE WORDS. Study the words and their definitions, and then fill in the sentences with correct words.

thermal management (phr) – controlling heat in computers to avoid damage or power waste.

energy audit (phr) – an analysis to find where and how to save energy in IT systems.

power cap (phr) – a limit set to reduce how much energy a server or device can use.

idle power draw (phr) – the energy a device uses when it's on but not doing tasks.

efficiency ratio (phr) – the amount of computing power compared to energy used.

low-power mode (phr) – a setting that reduces energy use by turning off non-essential functions.

server farm (phr) – a large group of servers often used by data centers.

on-demand computing (phr) – running systems only when needed to avoid wasting power.

zero-emission data center (phr) – a facility that runs fully on renewable or clean energy sources.

energy recovery (phr) – reusing waste heat or power generated during computing.

1. The company stores all its customer data in a large _____ outside the city.
2. Modern laptops use smart _____ to keep the processor cool without using too much energy.
3. The company built a _____ powered by wind and solar energy.
4. The data center set a _____ for each server to control energy costs during peak hours.
5. When I switch my laptop to _____, it reduces screen brightness and slows background apps.
6. Even when the printer is not being used, it has a high _____, so it's better to switch it off.
7. This new processor has a better _____, it performs more tasks while using less electricity.
8. Cloud providers use _____ to power up systems only when someone needs them.
9. After the _____, the IT team found that the office could save power by turning off unused monitors at night.
10. Some high-tech companies use _____ to heat their office buildings with waste heat from servers.

Task 26. WORK WITH WORDS. Read the text and fill in the gaps with the correct words.

Cutting IT Energy Waste

Modern data centers are working hard to become more energy-efficient. One of the key strategies is thermal management, which is keeping 1) _____ cool without wasting electricity. Instead of traditional air conditioners, many centers use advanced cooling systems that 2) _____ heat and improve safety.

Before making changes, companies often perform an energy audit to 3) _____ where power is being wasted. For example, servers that are running but not doing any work still use electricity. This is known as idle power draw. By using on-demand computing, companies can 4) _____ servers only when they are needed.

Some businesses apply a power cap, which sets a 5) _____ on how much electricity each server can use. To save even more, servers can be switched to low-power mode during 6) _____ hours. These methods help 7) _____ the efficiency ratio, meaning they get more computing power per unit of energy.

Large server farms, collections of 8) _____ servers, can generate a lot of heat. Instead of letting that heat go to waste, some centers use energy recovery systems to heat nearby buildings or water 9) _____.

The ultimate goal is to build a zero-emission data center that runs only on 10) _____ energy. With smart planning and new technology, the IT world is moving closer to that goal every year.

- | | | | | |
|-----|---------------|----------------|--------------|-------------|
| 1. | A) gear | B) tools | C) equipment | D) software |
| 2. | A) reduce | B) transfer | C) measure | D) trap |
| 3. | A) break down | B) look up | C) fix out | D) find out |
| 4. | A) fix | B) turn on | C) plug out | D) connect |
| 5. | A) deadline | B) bar | C) measure | D) limit |
| 6. | A) quiet | B) peak | C) noisy | D) day |
| 7. | A) delete | B) improve | C) install | D) repair |
| 8. | A) grouped | B) attached | C) connected | D) reserved |
| 9. | A) packages | B) tanks | C) storage | D) supplies |
| 10. | A) renewable | B) traditional | C) sun | D) digital |

Task 27. COMMUNICATE. Work in pairs. Discuss the questions below.

- Have you ever used low-power mode on your phone or laptop? Did it noticeably change performance or battery life?
- Why do companies build server farms in colder countries?
- Could energy recovery from computing be used to heat buildings in cold regions? What other ideas can you think of?
- What does a zero-emission data center need in order to work? Do you think it's realistic for every company?
- Should governments support more zero-emission data centers? Why or why not?
- Why is thermal management important in laptops, smartphones, or gaming computers? Have you ever had a device overheat?
- Imagine your university or office is doing an energy audit. What devices or behaviors do you think would use the most power?
- What are some situations where on-demand computing could help companies reduce power consumption?

Task 28. WORK WITH WORDS. Match power-saving modes to their explanations.

1. sleep mode (standby)	a) turns off the screen after inactivity
2. hibernate mode	b) saves the current state of the device to RAM and powers down non-essential components (screen, hard drive, etc.)
3. low-power mode (battery saver)	c) automatically lowers the CPU speed when full performance is not needed
4. power saver mode (Windows) / energy	d) completely powers down the device (printer, router etc.) after a period of inactivity

saver (macOS)	
5. display off	e) reduces screen brightness, background activity, and performance to extend battery life
6. CPU throttling	f) limits CPU performance, dims display, and reduces system activity
7. deep sleep	g) saves the current state to the hard drive and powers off the device completely

Task 29. COLLABORATE. Work in pairs. Read each situation and decide which power-saving mode would be most appropriate. Explain why you think so. Use the terms from the box.

sleep mode ▪ hibernate mode ▪ low-power mode ▪ display off ▪
power saver mode ▪ CPU throttling ▪ deep sleep ▪

1. Your smartphone is at 10% battery and you still have an hour until you get home.
2. Your desktop computer isn't doing any tasks, but it's still running. You don't want to shut it down completely.
3. You haven't used your home printer in 30 minutes. It turns itself off.
4. A cloud data center wants to reduce server energy use when demand is low at night.
5. You want your laptop battery to last longer during a flight, so you dim the screen and stop background apps.
6. You need to leave your laptop for several hours and want to save all your work while using no power.
7. You're stepping away from your laptop for 15 minutes during a lecture.

Task 30. COMMUNICATE. Work in pairs. Discuss the questions below.

1. Which power-saving mode do you use most often? Why?
2. Do you think people know enough about these options?
3. How can IT companies encourage better power-saving habits?
4. Have you ever had trouble waking a device from sleep or hibernate mode? What happened?

Task 31. EXPLORE THE IDIOMS. Study the meaning and examples of IT related idioms. Then use them in sentences.

Turn over a new leaf – to make a fresh, positive change.

*The company **turned over a new leaf** with energy-efficient servers.*

Drop in the bucket – a small effort compared to the problem.

*Recycling one device is a **drop in the bucket** for e-waste.*

Burn the candle at both ends – to overwork.

*Old computing methods **burn the candle at both ends**.*

Leave no stone unturned – to explore every option.

*They **left no stone unturned** to reduce power usage.*

Sow the seeds – to initiate.

*The policy **sowed the seeds** for greener computing.*

Run on fumes – to operate with minimal resources, often inefficiently.

*Old laptops **run on fumes**, draining power like crazy.*

Make a dent – to have a noticeable but limited impact on a big issue.

*Switching to LEDs **made a dent** in our energy bill, but more's needed.*

Fill in the gaps with the correct idiom from the list above. Each idiom is used only once. Sometimes you may need to change the form of the verb.

1. Fixing one computer is just a _____ compared to the piles of e-waste we have.
2. The team _____ by switching to solar power for their data center.
3. That old server _____, using way too much electricity all night.
4. His new idea _____ for a project to recycle phone batteries.
5. The engineers _____ to find a way to cut energy costs in the lab.
6. Better cooling _____ in our power use, but we still need bigger changes.
7. That slow machine _____, barely working and wasting power.

Task 32. COLLABORATE. Work in pairs. Imagine a space station crew stuck with old tech spilling e-waste into orbit. How do they clean it up? Use two or three idioms.

Task 33. WRITE. Write a short, funny story about a lazy tech geek who accidentally starts an e-waste revolution at home. Use the idioms above.

WATCHING

Task 34. WATCH FOR DETAILS. Scan the QR code and watch a video “Energy-efficient Data Centers”. Choose the correct answer for the questions.



1. What percentage of U.S. electricity is used by data centers?
 - A. 1%
 - B. 3%
 - C. 10%
 - D. 20%
2. What is a common issue with traditional data centers?
 - A. They use outdated servers.
 - B. They rely only on solar energy.
 - C. They run 24/7 and get very hot.
 - D. They are built only in cold climates.
3. How do energy-efficient servers save power?
 - A. They are turned off completely at night.
 - B. They run at full power all the time.
 - C. They switch to a low-power standby mode when not in use.
 - D. They only store a small amount of data.
4. What method is used in cold months at the efficient data center?
 - A. Ice blocks placed behind the servers.
 - B. Natural sunlight to keep servers warm.
 - C. Heat from servers is used to warm the building.
 - D. Cooling towers installed on the roof.
5. How is underground air used in this energy-efficient data center?
 - A. It's mixed with water for cooling.
 - B. It's sent through air conditioning units.
 - C. It is cooled and reused for other buildings.
 - D. It circulates through vents to cool the data center.
6. How often does the efficient data center use air conditioning?
 - A. Every day.
 - B. For 33 days each year.
 - C. Only when the temperature hits 100°F.
 - D. Just 33 hours per year.
7. Why is the climate important for energy-efficient data centers?
 - A. Warm climates help reduce electricity use.

- B. Cold climates reduce the need for mechanical cooling.
 - C. Hot climates support server performance.
 - D. Data centers must be near rivers.
8. How much energy does this energy-efficient data center use compared to a traditional one?
- A. 25% less.
 - B. About the same.
 - C. 81% less.
 - D. Twice as much.

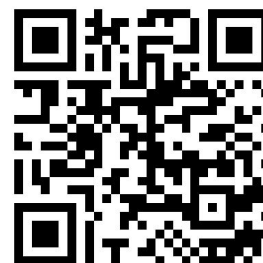
Task 35. COLLABORATE. Work in groups. Study the differences between traditional data centres and energy-efficient ones. Then discuss a situation when your company wants to make its data center greener but you have a limited budget. From this table, choose 3 features to upgrade. Explain your reasons.

Traditional vs. Energy-Efficient Data Centers

Feature	Traditional Data Center	Energy-Efficient Data Center
1. cooling system	always-on air conditioning	underground air / smart fans / reused hot air
2. server activity	servers run 24/7, even when idle	low-power standby mode when not in use
3. energy optimization	no energy-saving software	smart software manages energy use
4. heat waste	heat goes into the air	heat is reused to warm buildings
5. air circulation	basic ventilation	uses underground air at 12°C for cooling
6. energy use	uses 3x more electricity than equipment needs	uses 81% less energy than a traditional center

LISTENING

Task 36. LISTEN FOR DETAILS. Scan the QR code and listen to a podcast “Green Servers”. Answer the questions below.



Which data centre ...

1. uses cold sea water to keep its servers cool?
2. is fully powered by a large nearby solar farm producing 270 megawatts?
3. runs on hydropower and benefits from Nordic air for cooling?
4. is located in Finland and uses wind energy?
5. has a solar project called Gigawatt 1 to supply clean electricity?
6. is located in Wyoming and gets renewable power from wind projects?
7. has a massive campus called Citadel and is 100% solar-powered?
8. belongs to a company whose cloud services are powered by solar in Nevada?

- A. Apple
- B. Google
- C. Amazon Web Services
- D. Microsoft
- E. Switch

Task 37. COLLABORATE. Work in small groups. Choose one of the data centers mentioned in the podcast. Search for additional information about it on the Internet. Answer the questions below.

- What makes this data center energy-efficient?
- What type of renewable energy does it use?
- How does its location help reduce energy use?
- Why do you think this company made this choice?
- Would this model work well in your country or city? Why/why not?

Prepare a short 1-minute presentation to share your ideas with the class.

SPEAKING

Task 38. COLLABORATE. Work in small groups. Imagine your university wants to upgrade its computer lab using green hardware. Below are four devices your university can choose from. Read the descriptions and discuss your choice.

Eco Desktop Pro

- Energy Star-certified
- Recyclable aluminum casing
- High performance but uses a lot of power
- Moderate cost



GreenPrint Smart Printer



- Automatically reduces paper use
- Uses biodegradable ink cartridges
- Slower printing
- Low energy use

Modular Tablet



- Easy to repair and upgrade
- Made with recycled plastics
- Good battery life
- Medium performance

SolarBook Laptop

- Solar-powered with backup battery
- Lightweight and portable
- Low processing speed
- Expensive



Task 39. COMMUNICATE. Work in small groups. Discuss the questions below.

1. Would you choose one of these devices for your university? Why or why not?
2. Which device do you think is the most eco-friendly?
3. Which is the most practical for daily student use?
4. What are the advantages and disadvantages of using green hardware?
5. Do you think universities should always choose green hardware, even if it's more expensive?

Task 40. COMMUNICATE. Work in groups of four. Role play the situation below.

Your small IT company is reviewing its aging computer equipment. Some of the devices are still working but slow, and maintenance is becoming more frequent. The leadership is considering whether to upgrade the old computers or replace them with new green hardware. You are taking part in a team meeting to make a decision.

Student A: You are an IT specialist. You manage the computers. You want reliable and fast machines. You prefer high performance.

Student B: You are a Finance manager. You handle the budget. You're concerned about costs. You prefer upgrading if it saves money.

Student C: You are a Sustainability officer. You want the company to reduce its environmental footprint. You support investing in green tech.

Student D: You are an Operations manager. You care about workflow and downtime. You want a balance between cost, speed, and sustainability.

Task 41. COMMUNICATE. Work in pairs. Your company's electricity bills have increased sharply. Management asked your team to come up with ideas to reduce energy use in computing systems. Look at the list of ideas below. Choose at least three strategies that would help. Discuss the advantages and disadvantages of each.

- Enable sleep mode on all computers after 5 minutes of inactivity
- Replace old servers with modern energy-efficient models
- Move services to the cloud with lower carbon footprint
- Set up automatic power-down schedules during non-working hours
- Use energy monitoring software to track usage and find waste
- Encourage employees to unplug chargers and unused devices

Task 42. COLLABORATE. Divide into two teams to have a debate about energy efficiency in technology.

Some people say energy efficiency should always come first in computer design. Others think performance is more important. What do you think? What should come first in modern technology performance or energy efficiency?

Should computer designers focus more on performance or energy efficiency?

Performance	Energy efficiency
1. Powerful computers are needed for AI, research, gaming, and modern apps.	1. Energy-efficient tech reduces electricity use and helps the planet.
2.	2.

Task 43. ANALYZE. Read the case study and discuss possible solutions to the problem.

A tech firm based in Arizona operates a large data center that supports cloud computing services across the U.S. As summer temperatures reached record highs in 2025, the facility's air-based cooling systems began to fail. The cooling systems, mostly air-based, couldn't keep up anymore. By the time engineers started seeing patterns, several emergency shutdowns had already happened. They traced part of the issue to older server racks, units that still ran on outdated processors and inefficient power supplies. The hardware wasn't just old; it was running hot. The obvious fix? Upgrade the servers. But that would be expensive, and more importantly, it would take parts of the center offline, a risk the team wasn't sure they could afford. Now they're stuck trying to balance stability, cost, and uptime with a heatwave that's showing no signs of letting up.

What kinds of strategies (technical or operational) might help this company cool down its infrastructure and save energy without knocking services offline?

LANGUAGE FOCUS

Task 44. STUDY AND ANALYZE. Look at the rule about linking words, study in what situations they are used.

CLAUSES OF REASON

WORDS/EXPRESSIONS	EXAMPLE
because (usually answers a why-question).	Why is e-waste a problem? – Because electronic devices contain toxic materials.

for (always comes after a comma in writing)	They developed a new recycling method, for the previous one was ineffective.
because of / due to / on the account of + noun / -ing form	Because of poor e-waste management, landfills are filling up quickly. Due to using outdated hardware, the company's energy consumption increased.
due to the fact / because of the fact / on the account of + that-clause	Due to the fact that many people discard electronics improperly, pollution levels are rising.
since / as (=because)	Since energy-efficient computers consume less power, they help reduce electricity costs.
the reason for + noun / -ing form	The reason for using energy-efficient processors is to reduce power consumption.
the reason why+clause	<i>The reason why</i> energy-efficient computers are popular is that they save electricity and lower costs.
Now (that)+clause	Now that e-waste regulations are stricter, companies must follow recycling laws.

Task 45. PRACTICE. Circle the correct item.

1. **For** / **Since** electronic devices contain valuable metals, they should be recycled properly.
2. Many computers are discarded **because** / **due to** they become outdated quickly.
3. The factory switched to energy-efficient servers **as a result** / **for** of high electricity costs.
4. **For** / **As** e-waste can harm the environment, proper disposal is necessary.
5. Many old laptops end up in landfills, **therefore**, / **on the account that** recycling programs should be expanded.
6. The company upgraded its hardware **since** / **due to** the new energy efficiency policies.
7. **For** / **Since** I don't have an old phone to recycle, I'll donate my laptop instead.
8. The office installed LED screens **as** / **due to** they use less power.
9. **Because** / **Due to** the rapid growth of e-waste, new recycling centers are being built.
10. The company is focusing on energy-efficient solutions, **the reason why** / **for** they aim to reduce their carbon footprint.
11. **The reason for** / **why** using solar chargers **is** to save energy.

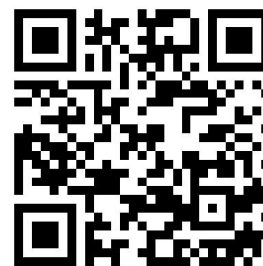
Task 46. PRACTICE. Join the sentences with the words in brackets, then identify the type of clause they introduce.

1. Many old electronics are discarded. They are difficult to repair and reuse. (as a result)
2. The new computer system uses less power. It can handle more tasks. (so ... that)
3. We recycle old phones and laptops. We reduce harmful e-waste. (for)
4. The energy-efficient processors were installed. The data center reduced its electricity consumption. (therefore)
5. E-waste disposal is a problem. People often throw old electronics in the trash. (on the grounds that)
6. Using cloud storage reduces hardware needs. It lowers energy consumption. (since)
7. E-waste recycling helps the environment. It recovers valuable materials. (as)
8. More companies are adopting green technologies. They want to improve sustainability. (due to the fact that)
9. Energy-efficient computing devices are in high demand. They help save money on energy bills. (due to)
10. The new recycling method reduces e-waste. It can recover more reusable components. (so ... that)

UNIT 7.3 CIRCULAR ECONOMY

WATCHING

Task 47. WATCH FOR DETAILS. Scan the QR code and watch a video “Circular economy”. Answer the questions below.



1. What is the main idea of the circular economy model?
2. How does the linear economy typically handle products after they are used?
3. Why is the light leasing example from Philips Lighting important?
4. What incentive does the circular economy give to companies like Philips?
5. How is H&M contributing to circular economy practices?
6. What year did the European Union adopt its circular economy action plan?
7. According to the video, what economic benefit could the circular economy bring to the EU by 2050?
8. What major challenge do manufacturers face in transitioning to a circular economy?
9. What percentage of UK companies are trying to get value from reused or recycled products?
10. What is one major obstacle (besides cost) to making the circular economy a reality?

Task 48. COLLABORATE. Work in pairs. Read the situations and decide which economy model each one refers to.

Linear Economy or Circular Economy?

1. Products are made, used, and then thrown away.
2. Companies stay responsible for the products even after selling or leasing them.
3. Focuses on using materials as if they are unlimited.
4. Products are collected, repaired, reused, or recycled.
5. The goal is to reduce waste and make better use of resources.
6. Customers buy and own physical products like clothes or electronics.
7. Requires new ways of thinking about design, supply chains, and ownership.
8. Includes recycling as part of the process.
9. The company has no further connection to the product after it is sold.
10. Helps both the environment and the economy over time.

Task 49. COMMUNICATE. Work in pairs. Discuss the questions below.

1. Which model do you think is better for the future?
2. Can you think of examples in your country where circular economy practices are used?
3. Would you prefer to lease products (like a phone or a car) instead of owning them? Why or why not?

READING AND VOCABULARY

Task 50. READ FOR DETAILS. Read the text and complete the sentences below.

Dell's Recycling and Refurbishment Facility

In the heart of Tennessee, USA, Dell has a cutting-edge facility in support of the circular economy. Instead of dumping old electronics, Dell accepts them and gives new life to materials. The facility is among Dell's Asset Recovery and Recycling Services. Laptops, desktops, and servers are shipped in from schools, businesses, and individuals. The facility handles thousands of devices weekly.

Initially, all items go through a data-erasing procedure to remove all personal information. Then, workers sort devices into two categories: products to be refurbished and products to be disassembled.

Refurbished laptops are cleaned, tested, and repaired. They are given new parts if required and then sold at a lower price as certified refurbished products. This helps to reduce e-waste and provide access to technology for more people.

For the ones that are impossible to repair, the plant carefully takes out parts like batteries, cables, and memory chips. Items like aluminum, plastic, and rare earth metals are stripped and exported to Dell's manufacturing partners to be used in new products. In some models, up to 35% of the plastic consists of recycled material.

Dell has a goal to recycle or reuse 100 million pounds (around 45 million kilograms) of used electronics every year. Dell also wants all its packaging to be recyclable or compostable by the year 2030.

The factory does not just process Dell machines; it accepts other electronics brands as well. This helps keep more devices out of landfills, no matter who made them. Workers follow strict environmental guidelines to dispose of hazardous materials, like lead and mercury, in a safe way so as not to harm people and the planet.

Dell also partners with schools and non-profits to donate recycled computers. This helps give students and under-served communities access to technology that they might otherwise be unable to afford. By extending the life of electronics, Dell bridges the digital divide while avoiding waste.

(adapted from <https://www.dell.com/en-us/dt/corporate/social-impact/advancing-sustainability/accelerating-the-circular-economy.htm>)

Use the correct word or number from the text to complete each sentence:

1. Dell wants to recycle or reuse _____ pounds of electronics each year.
2. The facility is located in the state of _____.
3. Up to _____ percent of plastic in some Dell models is made from recycled materials.
4. Old electronics come from businesses, individuals, and _____.
5. All devices are cleaned and _____ before being sold.
6. The facility also accepts electronics from other _____, not just Dell.
7. Hazardous materials like _____ and mercury are processed safely.
8. Donated computers help reduce the _____ divide.

Task 51. READ FOR DETAILS. Match each group or role to their action in the Dell recycling process.

1. Use recovered materials to make new products.	a) refurbishment team
2. Donate devices to people in need.	b) Dell company
3. Repair and test laptops for resale.	c) workers in disassembly
4. Recycle or reuse 100 million pounds of electronics per year.	d) schools and nonprofits
5. Remove parts like batteries and chips from broken devices.	e) manufacturing partners

Task 52. EXPLORE THE WORDS. Look at the words and study their definitions. Fill in the gaps in the sentences using some of these words.

product-as-a-service (n) – a business model where customers pay to use a product instead of owning it.

de-manufacturing (n) – carefully taking apart products to recover reusable or recyclable parts.

zero-waste (n) – a system that aims to eliminate all waste through reuse and efficient design.

downcycle (v) – to recycle materials into products of lower quality or value.

upcycle (v) – to transform waste materials into new products of higher value or quality.

closed-loop system (phr) – a production system where waste is collected and reused in the same cycle.

open-loop recycling (phr) – waste from one system is recycled and used in a different one.

digital product passport (phr) – a digital record that tracks the origin, use, materials, and recyclability of a product.

smart labeling (phr) – technology (like QR codes or chips) added to products to provide information about reuse, recycling, or repair.

modular design (phr) – building products with replaceable or upgradeable parts.

Now use them in the sentences below. Sometimes you need to change the form of the word.

1. In _____, plastic casings from old printers are turned into outdoor benches.
2. Many businesses now lease laptops through a _____ plan, which includes upgrades every two years.
3. The old laptops went through a _____ process to recover reusable parts like batteries and circuit boards.
4. Plastic from old keyboards was _____ into low-grade construction material.
5. Each smartphone now comes with a _____, showing where its parts come from and how to recycle them.
6. Some startups _____ e-waste, turning discarded computer parts into decorative items and furniture.
7. Thanks to _____, users can easily upgrade their computer's memory without replacing the entire device.
8. The company adopted a _____ policy to make sure that all packaging materials are reused or recycled.
9. The factory runs a _____ where all leftover materials are collected and reused in the next batch of production.
10. The monitor has _____ that lists all the materials used and how they should be handled at end-of-life.

Task 53. WORK WITH WORDS. Read a dialogue between two coworkers at a tech company discussing how their company is moving toward more sustainable production. Fill in the words from the box to fit the gaps.

product-as-a-service ▪ open-loop recycling ▪ smart labeling ▪ modular design
▪ closed-loop system ▪ de-manufacturing ▪ upcycle ▪ zero-waste ▪ downcycle
▪ digital product passport ▪

The New IT Model

Lena: Hi Tom, did you see the new laptop concept that our team working on sustainability

Tom: I saw the draft this morning. It uses 1) _____, right?

Lena: Yes! All parts, for example, the battery or the screen, can be replaced or even upgraded. Customers will not need to dispose of the whole product when one part is broken.

Tom: That makes us a 2) _____ company. Fewer broken machines in the dump.

Lena: And when the laptop is outdated, we will use 3) _____ to carefully take it apart. Then we can 4) _____ high-quality materials and 5) _____ the other parts into basic things like packaging.

Tom: I also read they want to move toward a 6) _____ for components where the same materials are reused again and again.

Lena: That's much better than 7) _____, where parts might be reused in completely different industries. It's harder to track where materials go.

Tom: Speaking of tracking, we're adding a 8) _____ to each device. It shows where every part came from and how it can be recycled.

Lena: And don't forget the 9) _____; you simply scan the QR code to receive repair

options or recycle data.

Tom: I adore how it all integrates into our 10) _____ plan. Customers will not need to buy the devices, they just pay for using them, and we take care of updates, repairs, and returns.

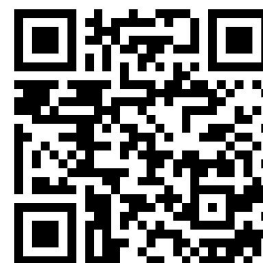
Lena: It's a win for the company, the users, and the planet.

LISTENING

Task 54. LISTEN FOR MAIN IDEA. Scan the QR code and listen to a podcast "Product-as-a-Service" and match each company mentioned in the podcast to the service it provides. One sentence is extra.

A. HP B. Vestas C. Grover

1. Offers wind turbines as a pay-per-use service, not a product sale.
2. Sends ink automatically and charges for pages printed.
3. Provides streaming music with subscription plans.
4. Lets people rent electronic devices like phones and laptops.



Task 55. LISTEN FOR DETAILS. Listen to the podcast again. Choose the correct answers to the questions.

1. HP Instant Ink sends you ink when...
 - A. your subscription ends.
 - B. the printer stops working.
 - C. you print more than 100 pages.
 - D. your printer runs low.
2. Grover allows customers to...
 - A. own a device after one year.
 - B. repair devices themselves.
 - C. rent devices and return or upgrade them.
 - D. buy electronics in bulk.
3. Vestas provides wind turbines as...
 - A. free equipment for cities.
 - B. a pay-per-use service.
 - C. permanent property for clients.
 - D. short-term experiments.
4. The "product-as-a-service" model helps reduce waste by...
 - A. keeping products in use longer.
 - B. lowering the cost of goods.
 - C. replacing parts every year.
 - D. using biodegradable packaging.
5. Customers like product-as-a-service because...
 - A. it increases energy use.
 - B. they avoid large upfront payments.
 - C. it requires no subscription.
 - D. devices come with no warranty.
6. One advantage for companies is that they...
 - A. avoid maintaining the products.
 - B. stop supporting old devices.
 - C. stay connected with the product even after it's used.
 - D. don't need to recycle anymore.
7. Grover's service is especially helpful for people who...
 - A. travel a lot and need new clothes.

- B. prefer in-store shopping.
- C. need high-tech gear temporarily.
- D. own multiple printers.

SPEAKING

Task 56. COMMUNICATE. Read the explanation of software obsolescence and discuss the questions below.

Software obsolescence happens when software is no longer supported, updated, or compatible with modern services even if the hardware is still working perfectly.

1. What happens when a phone or computer stops getting software updates?
2. Do you think companies stop updates too soon? Why or why not?
3. Have you ever had to replace a device just because the software stopped working? How did you feel about that?
4. Should companies be required to offer software support for a minimum number of years?
5. How can software obsolescence affect the environment?
6. Do you think software should be open for repair or updates after a company stops supporting it? Why or why not?
7. Would you buy a phone or laptop that promises 10 years of software updates?
8. Should governments create laws about software updates and device lifespans?

Task 57. COLLABORATE. Work in small groups. Shuffle the Device Cards below and place them face down. Take turns drawing one card, reading it aloud, and discussing with your group. Decide which category the device belongs to (**Still Useful /Needs Update /Should Be Recycled**) and explain your reasoning. Use support phrases if needed.

Useful phrases:

- The battery is weak, but the software works, so I think it's still useful.
- This could be upcycled if we replace....
- It's old and doesn't run modern apps so we should...
- This one only needs a software update so...
- Maybe someone else could reuse this device.

What's Worth Keeping?

CARD 1 Device Type: Laptop Age: 3 years Condition: Good Software Status: Up-to-date	CARD 2 Device Type: Smartphone Age: 6 years Condition: Cracked screen Software Status: No longer gets updates
CARD 3 Device Type: Desktop PC Age: 8 years Condition: Slow performance Software Status: Still running basic programs	CARD 4 Device Type: Smart watch Age: 1 year Condition: Excellent Software Status: Fully functional
CARD 5 Device Type: Tablet Age: 5 years Condition: Battery drains fast	CARD 6 Device Type: Printer Age: 10 years Condition: Working well

Software Status: Outdated apps	Software Status: No driver updates available
CARD 7 Device Type: Laptop Age: 4 years Condition: Broken keyboard Software Status: Software works fine	CARD 8 Device Type: Phone (basic) Age: 2 years Condition: Like new Software Status: Only runs basic apps
CARD 9 Device Type: Game console Age: 7 years Condition: Scratched body Software Status: Needs firmware update	CARD 10 Device Type: Router Age: 6 years Condition: Works intermittently Software Status: Security updates missing

Task 58. COLLABORATE. Work in small groups. During a tour of your university’s electronics recycling center, your group discovers a strange device in one of the bins. It doesn’t match any known brand. It has no label, but a soft green light is blinking. It’s made from ultra-thin metal and glass that don’t match current materials. The shape is unusual: part bracelet, part lens, part circuit. It glows brighter when someone is excited, emits a soft vibration when placed near metal. The technician says, “We’ve never seen anything like this.”



Discuss the gadget:

- What could it do?
- Who might have used it?
- Why was it discarded?
- Was it from the future?
- Was it an experimental prototype?
- Is it dangerous, valuable, or misunderstood?

Task 59. ANALYZE. Read the case study and discuss possible solutions to the problem.

TechCo is a medium-sized marketing company with 80 employees. They currently own all their computers, printers, and servers. Most of the equipment is 4–6 years old, often slow, and costly to repair. Energy use is also a concern. The IT manager has proposed switching to an electronic leasing model with a service provider. This would include:

- New laptops and printers delivered every 3 years
- Managed services: regular maintenance, remote updates, repairs
- Return and recycling of old equipment
- Fixed monthly payment per user

However, some company managers are hesitant. They worry about:

- Long-term costs of leasing
- Data security during equipment returns
- Losing flexibility in IT decisions

Discuss the following questions:

1. What are the current problems TechCo is facing?
2. What are the benefits of switching to a leasing model?
3. What are the possible drawbacks?
4. Should TechCo lease all devices?
5. Should they combine leasing and ownership (e.g., lease laptops, keep servers)?

WRITING

Task 60. WRITE. Imagine you are part of the university's sustainability committee or a tech student group. Write a formal suggestion or proposal (100–130 words) for an eco-friendly computing solution. Choose ONE idea from the list below or create your own:

- Create a used-device donation box in each campus building
- Organize a monthly e-waste drop-off event with certified recyclers
- Set up solar-powered charging stations for laptops and phones
- Develop a student app to track device lifespan and repair options

Your proposal should include:

- Clear purpose and context
- Concrete idea and how it works
- Benefits (environmental, financial, educational)
- Offer to meet or present more details

Useful phrases:

- I am writing to propose...
- We suggest implementing...
- This would reduce / improve / support...
- Our team is willing to...
- We would be glad to present a detailed plan...
- Thank you for considering this suggestion.
- Yours sincerely / Yours faithfully,

LANGUAGE FOCUS

Task 61. STUDY AND ANALYZE. Look at the rule about clauses of contrast and purpose, study in what situations they are used.

CLAUSES OF CONTRAST AND PURPOSE

WORDS/EXPRESSIONS	EXAMPLE
Clauses of contrast	
although / even though / though + clause	Even though tech companies design long-lasting products, many are still thrown away.
despite / in spite of + noun / -ing form	Despite using renewable energy, the tech industry creates a lot of pollution.
despite / in spite of the fact + that-clause	In spite of the fact that companies recycle e-waste, much of it is burned.
while / whereas / but / on the other hand / yet + clause	While energy-efficient gadgets save power, they are still expensive to make. The tech industry uses circular economy ideas, whereas consumers keep buying new gadgets.

	The new tech is energy-efficient, but it still produces a lot of waste. People want the latest tech, yet they ignore the need for recycling.
nevertheless / however + clause	The circular economy encourages recycling, however , some devices are hard to recycle.
however / no matter how + adjective / adverb + subject + (may) + verb	However eco-friendly their products are, some still end up as waste. No matter how green companies try to be, new tech creates a lot of waste.
whatever / No matter what + clause	Whatever efforts are made, tech products quickly become obsolete and are thrown away. No matter what companies do, it's hard to stop the growth of e-waste.
adjective / adverb + though + subject + verb / may + infinitive without to	Costly though renewable energy solutions are, they help the environment long-term.
adjective / adverb + as + subject + verb	Sustainable as the tech may be, its production still harms the environment.
Clauses of purpose	
to-infinitive	The company recycles old devices to reduce e-waste.
so as to / in order to + infinitive (formal)	He upgraded the server so as to improve its energy efficiency.
so that + can / may (present or future reference)	The company uses biodegradable materials so that it can minimize waste.
so that + could / might (past reference)	He recycled his old phone so that it might be used for parts.
for + noun (when we want to express the purpose of an action)	We use technology for improvement of energy efficiency in data centers.
for + -ing form (when we want to express the purpose or function of something)	This device is designed for minimizing electronic waste.
with a view to + -ing form	The government is investing in eco-friendly tech with a view to reducing pollution.
with the aim of + -ing form	We are implementing new recycling methods with the aim of reducing e-waste.
in case + present (present or future reference) / in case + past (past reference)	I'll carry my laptop in case I need to make any last-minute changes. They stored old equipment in case it could be recycled later.
for fear + might	He didn't throw away the old laptop for fear it might contain important data.

Task 62. PRACTICE. Circle the correct answer.

1. **However** / **Although** companies invest in recycling programs, e-waste continues to grow.
2. Many electronics are discarded **whereas** / **despite** they could be refurbished and reused.
3. **Despite** / **While** promoting sustainable production, some companies still rely on non-renewable materials.

4. Essential **whereas** / **as** recycling is, many consumers fail to separate their waste properly.
5. **Even though** / **Despite** the company adopted a circular economy model, some waste is still generated.
6. **In spite of** / **Whatever** efforts to reduce plastic use, packaging waste remains a challenge.
7. The tech company introduced a trade-in program. Customers were hesitant, **although** / **though**.
8. Some manufacturers use biodegradable components, **whereas** / **even though** others continue using toxic materials.
9. Efficient **though** / **but** e-waste management may be, illegal dumping remains an issue.
10. **Even though** / **Despite** sustainable policies, electronic waste levels remain high.

Task 63. PRACTICE. Fill in even though, however, whereas, though, or despite.

1. A: Did the company implement a recycling program?
B: Yes. They invested in eco-friendly waste management; _____, e-waste levels remain high.
2. A: Isn't this processor too old for modern applications?
B: Not at all. _____ its age, it still performs efficiently.
3. A: I think we use the same cloud storage provider.
B: No. Yours offers unlimited storage, _____ mine has a data limit.
4. A: The company reduced its plastic packaging.
B: I know. Their carbon emissions are still high, _____.
5. A: _____ refurbished laptops are cheaper, some people still prefer new ones.
B: True, but refurbishing helps reduce electronic waste.

Task 64. PRACTICE. Join the sentences using the words in brackets.

1. The company switched to renewable energy. It wanted to reduce its carbon footprint. (in order to)
2. The organization started a project to recycle old phones. They hoped to reduce the environmental impact. (with a view to)
3. The team recycled old electronics. They didn't want to contribute to e-waste. (so as not to)
4. They bought energy-efficient servers. They wanted to lower their power consumption. (to avoid)
5. The government implemented recycling policies. They aimed to reduce landfill waste. (with a view to)
6. The company installed a data backup system. Employees could recover files if necessary. (so that)
7. They provided spare chargers. They might lose theirs during travel. (in case)
8. The company used eco-friendly materials in production. They feared it would harm the environment if they didn't. (for fear that)
9. The company invested in solar panels. They wanted to reduce their energy consumption. (with a view to)
10. The company launched a new app. It was designed to track energy usage in real time. (for)